MACHINE LEARNING MINI-PROJECT · UE24CS352A

Bird Species Identification from an Image

Traditional machine learning versus EfficientNet-B0 transfer learning

Course	Machine Learning (UE24CS352A)
Dataset	CUB-200-2011
Methods	Handcrafted features + classic ML, and deep learning with transfer learning



Scope of the results

All reported results come from a balanced subset of CUB-200-2011: 4,000 training images (20 per species) and 400 test images (2 per species). They do not represent evaluation on the full 11,788-image dataset.

1. Abstract

Bird species identification from images is a challenging classification problem because different species can have similar colors, shapes and visual patterns. Manual identification requires expertise and can be time-consuming.

In this project we developed and compared two approaches on the CUB-200-2011 dataset. The first uses handcrafted image features with traditional machine learning algorithms. The second uses transfer learning with a pretrained EfficientNet-B0 convolutional neural network.

The traditional approach achieved **13.50%** test accuracy, while the fine-tuned EfficientNet-B0 approach achieved **55.00%** on the selected balanced dataset. In our experiments, transfer learning showed a clear advantage for fine-grained bird image classification.

13.50%	55.00%	+41.50
Method 1: RGB histogram + HOG + Linear SVM	Method 2: fine-tuned EfficientNet-B0	percentage points (55.00 – 13.50)

Scope of the results

All reported results come from a balanced subset of CUB-200-2011: 4,000 training images (20 per species) and 400 test images (2 per species). They do not represent evaluation on the full 11,788-image dataset.

2. Problem statement

The objective is to identify the species of a bird from an input image. The system takes a bird image as input, extracts or learns useful visual information, and predicts the corresponding species among **200** possible classes.

Main challenges

- Similar visual appearances between different bird species.
- Differences in lighting, background, pose and image quality.
- Fine-grained differences in color, shape and feather patterns.
- The large number of species to classify.

3. Dataset description

We used **CUB-200-2011**, a benchmark dataset for fine-grained bird classification. It contains 11,788 images of 200 bird species, with species labels and bounding-box annotations. Images vary in bird pose, background, lighting and appearance.

Balanced subset used in this project

Split	Images per species	Total images
Training	20	4,000
Test	2	400

The balanced subset lets both approaches be compared with the same number of training and test images per species.

> The reported results come from this 4,000 / 400 image subset. They should not be read as performance on the full dataset.

4. Overall methodology

We implemented two approaches on the same balanced subset. Throughout this report, **amber** marks Method 1 and **teal** marks Method 2.

METHOD 1
Traditional machine learning

1. Preprocess and crop bird images
2. Resize images to 128 × 128 pixels
3. Extract handcrafted image features
4. Scale the feature vectors
5. Train and evaluate traditional classifiers
6. Compare feature combinations and models

METHOD 2
Deep learning with transfer learning

1. Load a pretrained EfficientNet-B0 model
2. Replace the final layer with 200 outputs
3. Train the new layer; keep the pretrained feature extractor frozen
4. Unfreeze the last two feature blocks and fine-tune
5. Evaluate the model and generate sample predictions

METHOD 1 · TRADITIONAL MACHINE LEARNING

5. Method 1: handcrafted features

5.1 Image preprocessing

We used the bird bounding-box annotations to focus on the bird region. The cropped images were resized to 128 × 128 pixels before feature extraction. This standardized the image size and focused classification on the bird.

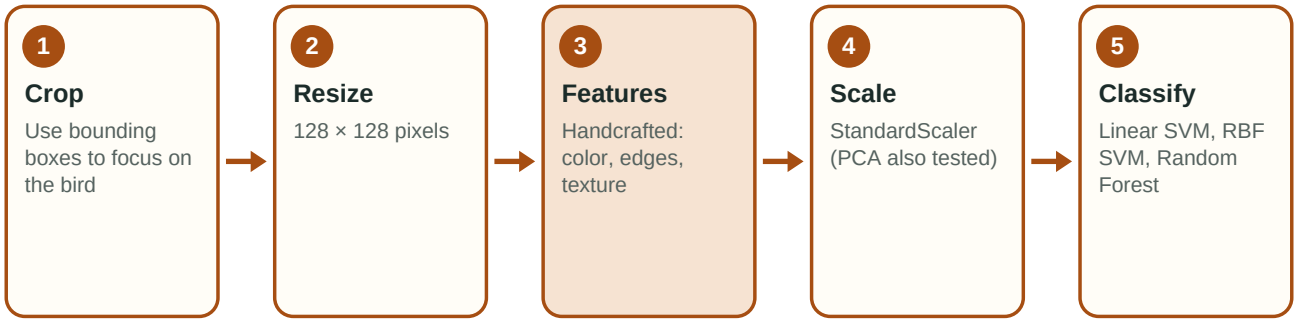


Figure 1. Method 1 pipeline.

5.2 Handcrafted feature extraction

Feature	What it captures
RGB color histogram	Distribution of red, green and blue values: the bird's overall colors (96 features).
HOG (Histogram of Oriented Gradients)	Local gradient directions and edge patterns: shapes, contours and structure (1,764 features).
LBP (Local Binary Patterns)	Local texture, by comparing each pixel with its neighbors: feather texture.
HSV color features	Color as hue, saturation and value; we explored spatial HSV from different image regions.
Color moments	Mean, standard deviation and skewness of the color distribution.
Gabor features	Directional and frequency-based texture: repeated patterns.

5.3 Feature combinations

- RGB Color Histogram + HOG
- RGB Color Histogram + HOG + LBP
- Spatial HSV + HOG
- Color Moments + HOG
- Gabor Features + HOG + RGB features

The RGB color histogram has 96 features and HOG has 1,764, so their combination gives **1,860 features** per image before scaling.

5.4 Feature scaling and PCA

We used StandardScaler to standardize the feature values before training. We also experimented with Principal Component Analysis (PCA) to reduce dimensionality while preserving important information: in one experiment PCA reduced the 1,860 features to 504 components, retaining about 95.02% of the variance.

5.5 Models

We evaluated a Linear Support Vector Machine (Linear SVM), a Radial Basis Function SVM (RBF SVM) and a Random Forest, across different feature representations.

5.6 Results

The best reported result was obtained with RGB Color Histogram + HOG features and a Linear SVM: **13.50%** test accuracy, which is 54 correct predictions out of 400 test images.

Experiment	Test accuracy
RGB + HOG + Linear SVM	**13.50%**
RGB + HOG + LBP + Linear SVM	13.25%
PCA + Linear SVM	12.50%
RGB + HOG + Random Forest	9.00%
RGB + HOG + RBF SVM	6.50%
Color Moments + HOG	6.50%
Spatial HSV + HOG	6.25%
Gabor + HOG + RGB	6.00%

Table 1. Method 1 results on the 400 test images (best result highlighted).

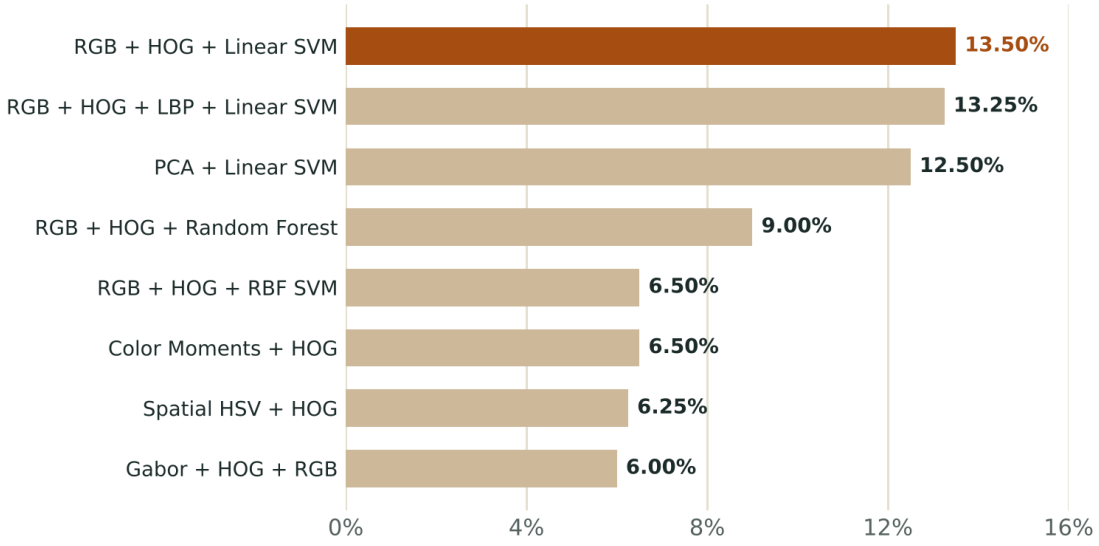


Figure 2. Method 1 test accuracy by experiment.

5.7 Limitations of Method 1

Handcrafted features represent selected visual properties, but may not capture the complex differences between closely related bird species. They can be sensitive to pose, background, lighting and changes in the bird's position, and the large number of classes makes accurate classification difficult. These limitations motivated us to try transfer learning with a pretrained deep learning model.

METHOD 2 · EFFICIENTNET-B0 TRANSFER LEARNING

6. Method 2: transfer learning

6.1 Key ideas in plain language

CNN	A Convolutional Neural Network is a deep learning model designed for images. It learns visual patterns by itself: early layers pick up simple patterns such as edges and textures, deeper layers pick up more complex structures. Unlike traditional machine learning, we do not design the features by hand. *Like a bird-watcher who learns what to look for.*
Transfer learning	Reuse knowledge a model learned on one dataset for a different task. We loaded pretrained EfficientNet-B0 weights and replaced the final layer with a new one of 200 outputs, one per species. *Like a student who can already read, starting a new subject.*
Fine-tuning	After the new layer is trained, unfreeze part of the pretrained model and train it gently so it adapts to birds. We unfroze the last two feature blocks. *Like adjusting an instrument you already know how to play.*

The analogies are teaching aids, not project results.

6.2 Why EfficientNet-B0?

EfficientNet-B0 is a convolutional neural network architecture designed to balance model size, computational requirements and classification performance. We chose a pretrained model so it could reuse visual features learned from a large image dataset instead of learning every visual pattern from scratch.

6.3 Preprocessing and pipeline

We used the preprocessing transformations associated with the pretrained EfficientNet-B0 weights: images are resized and normalized to the model's expected input format, at **224 × 224** pixels. The same 4,000 training and 400 test images across 200 species were used.

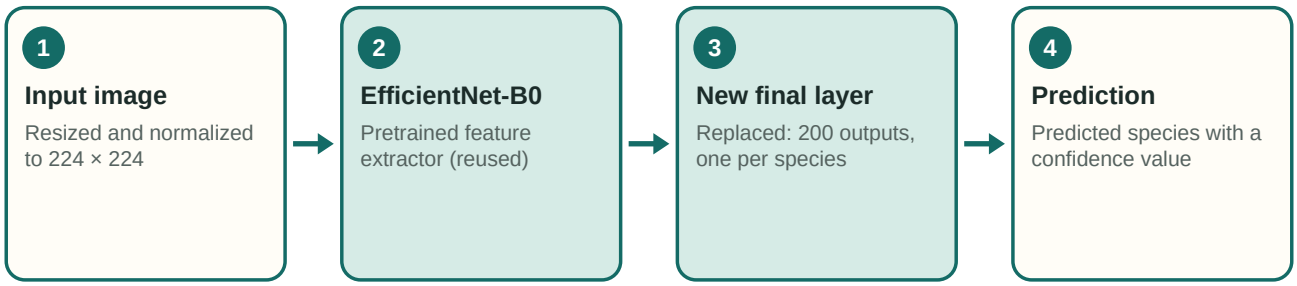


Figure 3. Method 2 pipeline. Shaded boxes: pretrained part and newly added layer.

6.4 Stage 1 and Stage 2 training

Stage 1. We froze the pretrained feature-extraction layers and trained the new classification layer for three epochs. **Stage 2.** We unfroze the last two feature blocks of EfficientNet-B0 so part of the pretrained extractor could adapt to bird species, and trained three additional epochs using the Adam optimizer with a learning rate of 0.0001.

Stage	What was trained	Training accuracy (epochs 1, 2, 3)	Test accuracy
Stage 1	New classification layer; feature extractor frozen	13.82%, 59.00%, 73.92%	50.25%
Stage 2	Last two feature blocks unfrozen (Adam, learning rate 0.0001)	82.55%, 85.55%, 87.97%	**55.00%**

Table 2. Method 2 training and test results.

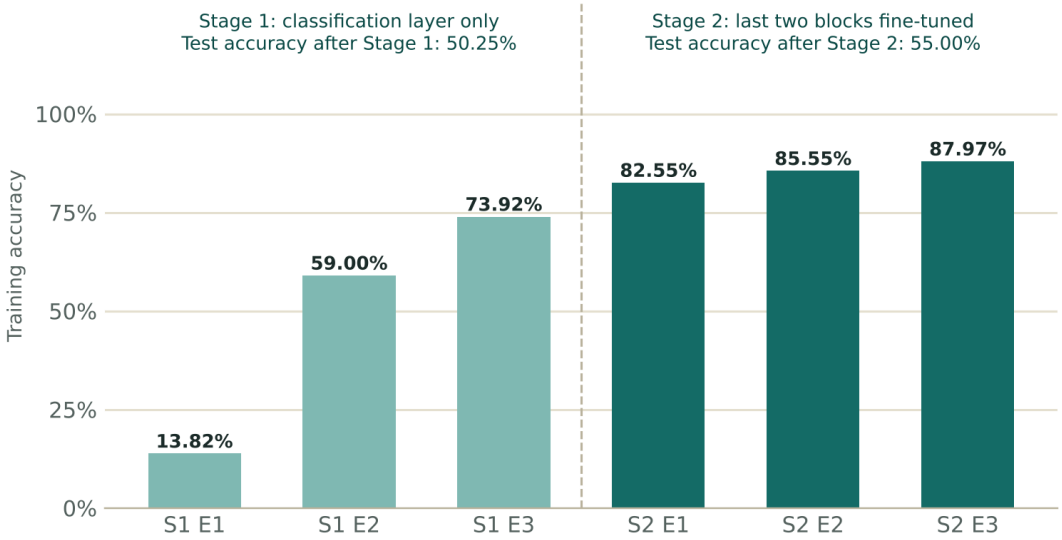


Figure 4. Training accuracy by epoch (S = stage, E = epoch). Test accuracy is measured on the 400 test images.

6.5 Prediction and confidence

The trained model produces class scores for an input image, and the predicted species is chosen from them. The notebook also reports a confidence value for sample predictions. We used the model on individual images and on a sample of ten test images, which demonstrates the full workflow from an input image to a predicted species. The trained model state is saved as `efficientnet_b0_cub_finetuned.pth`.

7. Comparison of both methods

Criteria	Method 1	Method 2
Approach	Traditional machine learning	Deep learning and transfer learning
Feature extraction	Handcrafted features	Features learned by EfficientNet-B0
Main representation	RGB histogram + HOG	Pretrained CNN representations
Main classifier / model	Linear SVM	Fine-tuned EfficientNet-B0
Input image size	128 × 128	224 × 224
Training images	4,000	4,000
Test images	400	400
Best reported test accuracy	**13.50%**	**55.00%**

Table 3. Side-by-side comparison.

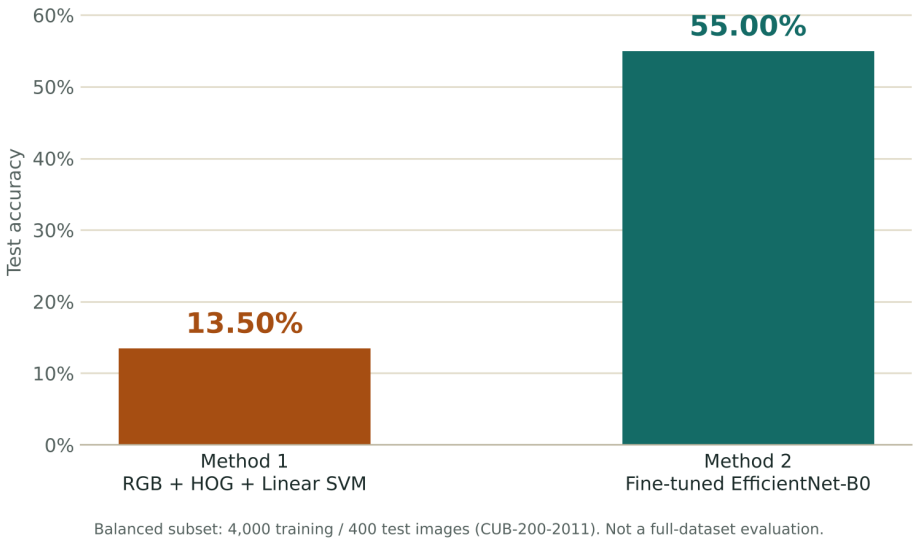


Figure 5. Best reported test accuracy of each method.

Accuracy improvement

Method 1 achieved 13.50% test accuracy and Method 2 achieved 55.00%. The absolute improvement is 55.00 – 13.50 = **41.50 percentage points**. The fine-tuned EfficientNet-B0 model performed better than the traditional approach in our experiments, which suggests that pretrained deep learning representations were more effective than the handcrafted features we tested for this fine-grained classification task.

Scope of the results

All reported results come from a balanced subset of CUB-200-2011: 4,000 training images (20 per species) and 400 test images (2 per species). They do not represent evaluation on the full 11,788-image dataset.

8. Implementation and tools

The project was implemented in Python using Jupyter Notebook, with separate notebooks for the two methods so that each implementation and its results can be examined independently.

Tool	Used for
Python	Implementation
NumPy	Numerical operations
pandas	Data handling, where applicable
OpenCV	Image processing and feature extraction
scikit-learn	Feature scaling, PCA and traditional machine learning models
PyTorch and Torchvision	EfficientNet-B0 transfer learning
Matplotlib	Visualization, where used

9. Conclusion

This project explored two approaches for identifying bird species from images. The first used handcrafted features (RGB color histograms, HOG, LBP, HSV, color moments and Gabor features) with traditional classifiers; its best reported test accuracy was **13.50%**, using RGB color histograms, HOG and a Linear SVM.

The second used transfer learning with pretrained EfficientNet-B0. We first trained the new classification layer and then fine-tuned the last two feature blocks, reaching **55.00%** test accuracy. Transfer learning performed substantially better than the handcrafted-feature approach in our experiments.

Overall, the project gave practical experience in image preprocessing, handcrafted feature extraction, traditional machine learning, convolutional neural networks, transfer learning, fine-tuning and model evaluation.

Future improvements

- Use a larger training subset.
- Apply data augmentation.
- Tune hyperparameters.
- Evaluate additional architectures.
- Examine per-species performance using confusion matrices and other classification metrics.

These have not been run, so no claims are made about their effect.

10. Repository and reproducibility

The project is organized into two notebooks: (1) Traditional Machine Learning and (2) EfficientNet-B0 Transfer Learning. The GitHub repository should contain the notebooks, a README file, a dependency list and the project report. The trained model file may also be included if available and appropriate for the repository size. The CUB-200-2011 dataset should be obtained separately rather than uploaded to GitHub. The notebooks preserve the experiments and reported outputs used to document the project.

Appendix: glossary for beginners

Term	Plain-language meaning
Test accuracy	Share of test images whose species was predicted correctly. For example, 13.50% is 54 correct out of 400.
Training / test images	Training images teach the model; test images check how well it does on pictures it was not trained on.
Fine-grained classification	Telling apart categories that look very similar, such as closely related bird species.
Handcrafted feature	A measurement of an image (such as color counts or edge directions) chosen by a person.
Epoch	One full pass through the training images.
Frozen layers	Parts of a model whose weights are not changed during training.
Confidence	A score showing how strongly the model favors its predicted species.
PCA	A method that compresses many features into fewer, keeping most of the information.
SVM	A classifier that separates classes with a boundary learned from the training data.